



SAJDA

Islamic Community Platform — Empowering the Ummah Digitally

"Berlumba-lumbalah kamu kepada kebaikan. Di mana saja kamu berada, pasti Allah akan mengumpulkan kamu sekalian. Sesungguhnya Allah Maha Kuasa atas segala sesuatu."

(Surah Al-Baqarah 2:148)

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1. Executive Summary

SAJDA is a comprehensive Islamic Community Platform designed to digitally empower mosques, their administrators, and the Muslim community across Malaysia and Southeast Asia. Built by a Malaysian Muslim founder who understands the *deen* from within, SAJDA bridges the gap between the physical masjid and its community members through technology — making Islamic practice more accessible, more consistent, and more connected.

The platform operates on a multi-tenant SaaS model: mosques subscribe to SAJDA as institutions, gaining access to a full administrative suite covering prayer schedule management, imam and bilal jadual, program registration, announcements, and community governance. Community members follow their registered mosques, participate in Islamic challenges, track their ibadah, join usrah groups, and contribute sadaqah digitally — all within a single, purpose-built Islamic digital environment.

SAJDA's competitive moat is unique: it is the only platform that connects the physical mosque infrastructure (AJK administration, imam schedules, real jemaah attendance) with a digital community layer featuring Islamic-context gamification. This integration — between the authoritative institution of the masjid and the daily spiritual life of its jemaah — cannot be replicated by generic community apps or prayer-time utilities.

The platform is built on Next.js 14, Supabase (PostgreSQL with Row Level Security), deployed on Vercel, with AI capabilities via Claude API. Billing is handled through Billplz for FPX and credit card support. The design system follows the 'Sacred Futurism / Dark Sanctuary' aesthetic — sacred green (#3DFF8F) on near-black backgrounds — designed to feel dignified, modern, and distinctly Islamic.

2. Introduction

2.1 Background and Motivation

Malaysia has approximately 6,500 registered mosques and surau, serving a Muslim population of over 21 million. Despite being central to Malaysian Muslim life — for prayer, education, community welfare, and spiritual development — the vast majority of these institutions operate with minimal digital infrastructure. Administrative tasks are managed through WhatsApp groups, paper notice boards, and verbal announcements. Attendance is estimated, not tracked. Programs are announced but not managed. Community needs go unmet because there is no platform to surface and connect them.

Meanwhile, Malaysian Muslim youth are increasingly disengaged from the mosque. A 2023 study by YADIM found that fewer than 30% of Muslim youth aged 18–35 attend Friday prayers regularly, and fewer than 15% attend weekday jemaah. The reasons cited are consistent: mosques feel disconnected from their daily digital lives, there is no meaningful engagement beyond solat times, and the sense of belonging that the early Muslim community felt in their masjid has been lost in the modern urban context.

SAJDA was conceived during the RC26 Hackathon, built for Masjid Saujana Utama in Bandar Saujana Utama, Sungai Buloh. The hackathon prototype proved the concept: a mosque management system with community features can meaningfully increase engagement. This SRS document defines the full commercial product vision that grows from that prototype into a platform for the global ummah.

2.2 Vision Statement

"To be the digital heartbeat of every mosque in the world — the platform through which the ummah learns, grows, gives, connects, and comes alive."

2.3 Scope

SAJDA v1.0 encompasses the following systems within scope:

- Mosque Management Suite — prayer schedules, imam/bilal jadual, program management, announcements, keperluan komuniti board, volunteer slots, kelas booking, AJK admin dashboard.
- Community Member PWA — mosque following, prayer times, community feed, ibadah tracking, gamification (Istiqamah Meter, Ilmu Quest, Khatam Together), doa network, usrah management.
- Gamification Engine — Islamic-context streak system, community challenges, Ramadan Mode, badges, and leaderboards designed with ikhlas-first principles.
- Digital Sadaqah Wallet — mosque-level digital donation collection with full transparency reporting and 1.5% transaction-based revenue model.
- Qiblat Finder — device sensor-based qibla direction.
- Hadis Harian — daily hadith from 40 Hadith Nawawi via fawazahmed0/hadith-api.
- Multi-tenant SaaS billing — three-tier subscription model (Surau, Kariah, Komuniti) via Billplz.

Out of scope for v1.0: WhatsApp bot integration (Phase 2), Islamic microfinance/qardhul hasan (Phase 3), waqf management module (Phase 3), Android native app (Phase 2 via PWABuilder).

2.4 Definitions and Acronyms

Term	Definition
AJK	Ahli Jawatankuasa — mosque committee members responsible for administration
Jemaah	Congregation — the mosque's registered community of worshippers
Imam	Prayer leader — leads the solat jemaah for the mosque
Bilal	Muezzin — calls the azan and assists the imam in mosque operations
Usrah	Small Islamic study/tarbiyah group, typically 5–12 members with a naqib/naqibah leader
Khatam	Completion of a full Quran recitation from Al-Fatihah to An-Nas
Istiqamah	Consistency and steadfastness in religious practice
Sadaqah	Voluntary charitable giving, distinct from the obligatory zakat
Zakat	Obligatory annual wealth purification payment, one of the five pillars of Islam
Waqf	Islamic endowment — property or assets donated for religious/charitable purposes
Ikhlas	Sincerity of intention — doing ibadah solely for Allah, not for show
Riak	Showing off — performing ibadah to be seen by others, a major sin
Muhasabah	Self-accountability and reflection on one's deeds
Naqib/Naqibah	Male/female usrah group leader
PWA	Progressive Web Application — web app with native capabilities (camera, offline, installable)
RLS	Row Level Security — Supabase/PostgreSQL data isolation mechanism
MRR	Monthly Recurring Revenue
JAKIM	Jabatan Kemajuan Islam Malaysia — Department of Islamic Development Malaysia
MAIS	Majlis Agama Islam Selangor — Selangor Islamic Religious Council

3. Problem Statement

3.1 The Digital Gap in Malaysian Mosque Management

The Malaysian mosque faces a structural digital gap that manifests across three dimensions: administrative inefficiency, community disengagement, and missed impact potential.

3.1.1 Administrative Inefficiency

Most Malaysian mosques manage their operations through a fragmented set of tools: WhatsApp for communication, paper ledgers for attendance, verbal announcements for programs, and memory for imam rotation schedules. This creates four measurable problems: information reaches jemaah inconsistently and late; AJK members spend hours on administrative tasks that could be automated; imam and bilal scheduling conflicts occur frequently due to lack of a shared digital calendar; and program attendance is untracked, making it impossible to evaluate program impact or identify community needs.

3.1.2 Youth Disengagement

The generation that spends 6+ hours daily on TikTok, Instagram, and Discord receives zero meaningful digital touchpoints from their mosque. There is no app notification reminding them of Subuh jemaah, no streak rewarding their consistency, no peer community celebrating their Quran reading progress, and no social proof that their friends are also attending the mosque. The mosque exists in an entirely different digital universe from where Malaysian youth actually spend their attention.

3.1.3 Missed Community Impact

Mosques are the original community welfare organisations in Islam — the Prophet SAW's first act upon arriving in Madinah was to build a masjid that served as school, court, community centre, and shelter. Today, mosques have enormous latent capacity to connect community needs (a single mother needing groceries, an unemployed youth needing job connections, an elderly person needing medical accompaniment) with community resources (volunteers, professionals, donors). This matching function is entirely absent from current mosque operations because there is no digital infrastructure to facilitate it.

3.2 Gap Analysis — Current vs Target State

Problem	Current State	SAJDA Target State
Program announcements	Verbally or WhatsApp group; many jemaah miss them	App notification + registration; full attendance tracking
Imam/bilal scheduling	Verbal rotation, frequent conflicts and confusion	Digital jadual with automatic conflict detection and notifications
Prayer times	Third-party apps with no mosque connection	Live prayer times from the subscribed mosque, with imam name displayed
Community needs	Unmet — no platform to surface them	Keperluan Komuniti board with volunteer matching
Youth engagement	Near zero — no digital touchpoints	Gamified ibadah tracking, peer challenges, community streaks
Sadaqah collection	Physical donation boxes; no tracking or reporting	Digital wallet with real-time collection dashboard and impact reports
Quran learning	Individual, disconnected	Community Khatam challenges with collective progress tracking
Jemaah attendance	Estimated by visual count	QR check-in with statistical dashboard for AJK
Kelas booking	Manual registration on paper or WhatsApp	Online booking with capacity management and automated reminders
Community welfare	Individual informal charity	Structured Keperluan Komuniti board with verified needs

4. Project Objectives

4.1 Primary Objectives

1. Deliver a complete mosque administration suite that replaces WhatsApp-based management for AJK, reducing administrative overhead by at least 60% for participating mosques.
2. Build a community member PWA that creates a meaningful digital connection between jemaah and their mosque — prayer times, programs, community feed, and ibadah tracking in one place.
3. Implement an Islamic-context gamification system that increases weekly mosque attendance and daily ibadah consistency among youth members, measurable through streak analytics.
4. Launch a digital sadaqah collection system that increases mosque donation collection volume by enabling anytime, anywhere digital giving via FPX and card.
5. Reach 100 subscribed mosques and 25,000 active community members within 12 months of commercial launch.
6. Achieve RM15,000 Monthly Recurring Revenue within 12 months through subscription fees and sadaqah transaction fees.
7. Expand to Indonesia and serve 1,000 mosques across Malaysia and Indonesia within 3 years.

4.2 Technical Objectives

1. Implement a multi-tenant Next.js 14 application where each mosque's data is fully isolated via Supabase Row Level Security, supporting unlimited mosque tenants on a single infrastructure.
2. Achieve sub-2-second page load for the community member PWA on 4G mobile connections — the primary access method for Malaysian mosque members.
3. Design the gamification engine with ikhlas-first principles: all individual ibadah tracking is private by default; only collective community achievements are surfaced publicly.
4. Integrate Claude API for AI-powered features: Hadis context explanation, Ilmu Quest question generation, and program description drafting for AJK.
5. Build the Ramadan Mode feature to be activatable per-mosque by AJK, launching automatically on 1 Ramadan and deactivating on 1 Syawal.
6. Achieve 99.5% uptime for prayer time display — this is safety-critical for the product's core value proposition.

5. Stakeholder Analysis

Stakeholder	Role	Primary Need	Success Metric
Masjid (Committee) AJK	Primary B2B customer; admin user	Replace WhatsApp chaos; manage programs, imam jadual, announcements professionally	Saves >5 hours/week on admin tasks; program attendance increases 30%
Imam & Bilal	System user — jadual management	Know their schedule in advance; receive reminders; manage leave	Zero scheduling conflicts; receive 48-hour advance notifications
Jemaah (adult members)	Consumer user — community follower	Know what's happening at their mosque; connect with community	Logs in weekly; attends at least 1 program/month through app booking
Muslim youth (18–35)	Primary gamification target	Feel connected to mosque through familiar digital experience	Weekly active user; maintains 7+ day ibadah streak
Mosque Naqib/Naqibah	Usrah group leaders	Manage their usrah group, track member attendance and amal	Weekly usrah meeting logged; all members active in app
Donors & sadaqah givers	Sadaqah wallet users	Easy, transparent, impactful giving	Gives digitally at least once per month; sees impact report
JAKIM / MAIS	Regulatory stakeholder	Mosques operating professionally and transparently	Supports or endorses SAJDA; no compliance issues
Founder (Danish)	Product owner & engineer	Build a sustainable, mission-driven Islamic platform	RM15K MRR by month 12; 100 subscribing mosques

6. System Features

6.1 Feature Matrix — Complete System Overview

Feature	Module	User	Tier	Priority
Prayer times (solat schedule)	Mosque Mgmt	All users	All tiers	Must Have
Imam & bilal jadual management	Mosque Mgmt	AJK, Imam, Bilal	All tiers	Must Have
Program/event management & booking	Mosque Mgmt	AJK + Jemaah	All tiers	Must Have
Announcement publishing	Mosque Mgmt	AJK	All tiers	Must Have
Keperluan Komuniti board	Mosque Mgmt	AJK + Jemaah	Kariah+	Must Have
Volunteer slot management	Mosque Mgmt	AJK + Jemaah	Kariah+	Must Have
Kelas booking system	Mosque Mgmt	AJK + Jemaah	All tiers	Must Have
AJK admin dashboard	Mosque Mgmt	AJK	All tiers	Must Have
Jemaah QR attendance check-in	Mosque Mgmt	Jemaah	Kariah+	Must Have
Follow mosque (subscribe)	Community	Jemaah	All tiers	Must Have
Community feed — Papan Komuniti	Community	AJK + Jemaah	Kariah+	Must Have
Doa & prayer request network	Community	Jemaah	Kariah+	Must Have
Usrah group management	Community	Naqib + Jemaah	Komuniti	Must Have
Islamic mentor matching — Cari Ilmu	Community	Jemaah	Komuniti	Should Have
Cross-mosque challenge network	Community	All	Komuniti	Should Have
Istiqamah Meter — solat streak	Gamification	Jemaah	Kariah+	Must Have
Khatam Together — Quran challenge	Gamification	Jemaah	Kariah+	Must Have
Ilmu Quest — daily Islamic knowledge	Gamification	Jemaah	Kariah+	Must Have
Amal Harian — daily deeds checklist	Gamification	Jemaah	Kariah+	Must Have

Feature	Module	User	Tier	Priority
Jemaah Points — masjid check-in	Gamification	Jemaah	Kariah+	Must Have
Kebun Amal — anonymous sadaqah board	Gamification	Jemaah	Kariah+	Must Have
Ramadan Mode — 30-day challenge	Gamification	All	Kariah+	Must Have
Islamic badges & achievements	Gamification	Jemaah	Kariah+	Should Have
Qiblat Finder	Tools	Jemaah	All tiers	Must Have
Hadis Harian — daily hadith + AI context	Tools	Jemaah	All tiers	Must Have
Digital sadaqah wallet	Sadaqah	AJK + Jemaah	Kariah+	Must Have
Sadaqah impact reporting	Sadaqah	AJK + Jemaah	Kariah+	Must Have
Zakat calculator (informational)	Tools	Jemaah	All tiers	Should Have

6.2 Detailed Feature Descriptions

6.2.1 Mosque Following System

Any registered Muslim user can search for and follow mosques that are subscribed to SAJDA. Following a mosque grants the user access to that mosque's prayer times, programs, community feed, and gamification challenges. A user may follow multiple mosques — for example, their home mosque and their workplace mosque. The primary mosque (set by the user) determines which prayer times appear on the home screen and which Ramadan Mode challenge they participate in. Mosques can be searched by name, location, or state. The following relationship creates the data link that allows SAJDA to personalise the user experience around their actual mosque community rather than generic Islamic content.

6.2.2 Imam & Bilal Jadual Management

The AJK admin creates and manages a rotation schedule for imams and bilals across all five daily prayers and Friday prayers. The system supports: weekly rotation templates that auto-generate schedules, individual prayer assignment overrides, leave/replacement requests from imams that trigger automatic notifications to the next assigned imam, and a shared calendar view accessible to all assigned imams and bilals. Imams receive push notifications 2 hours before their assigned prayer. The AJK dashboard shows the full month's schedule at a glance with conflict highlighting. Historical attendance records are maintained for accountability reporting to MAIS/JAKIM.

6.2.3 Program Management & Kelas Booking

AJK creates programs (ceramah, usrah, kelas Quran, gotong-royong, sports events) with full metadata: title, description, date/time, venue, capacity, speaker/facilitator name, program type, and whether registration is required. Jemaah browse programs from their followed mosques and register with one tap. The system manages capacity: when full, a waitlist activates automatically. Registered attendees receive reminder notifications 24 hours and 1 hour before the program. Post-program, AJK can mark attendance and generate attendance reports. AI (Claude API) assists AJK in drafting program descriptions from brief inputs.

6.2.4 Keperluan Komuniti Board

A structured board for surfacing and matching community needs with community resources. Categories: Bantuan Makanan (food assistance), Bantuan Kewangan (financial aid), Bantuan Perubatan (medical accompaniment), Keperluan Rumah (household needs), Kerja/Peluang (job opportunities), Lain-lain. Submissions are reviewed by AJK before publishing. The person in need does not reveal their identity publicly — AJK acts as the privacy intermediary. Community members who can help contact AJK. The board creates a structured, dignified welfare function that replaces informal and potentially embarrassing personal requests.

6.2.5 Digital Sadaqah Wallet

Each subscribed mosque has a dedicated digital sadaqah wallet accessible to its jemaah. Members contribute via FPX (online banking) or credit/debit card through Billplz integration. AJK sets up allocation categories: Tabung Masjid, Bantuan Fakir Miskin, Biasiswa Pendidikan, Bantuan Bencana, and custom categories. All collections and disbursements are logged with timestamps and purpose notes. Monthly impact reports are auto-generated showing total collected, total disbursed by category, and remaining balance. Reports are visible to all jemaah members of that mosque — full transparency. SAJDA charges 1.5% transaction fee on all sadaqah collections, capped at RM15 per transaction.

6.3 Gamification Engine — Ikhlas-First Design Philosophy

Design Principle: Every gamification element in SAJDA must pass the Ikhlas Test — "Does this feature encourage the user to remember Allah, or does it encourage the user to perform for others?" All individual ibadah tracking is private by default. Collective achievements are visible. The goal is to use social accountability to encourage istiqamah, not to create a platform for spiritual pride (riak).

Feature	Mechanic	Islamic Parallel	Privacy
Istiqamah Meter	Daily solat completion streak. 5 solat = 1 complete day. Visual garden grows with streak. Resets on miss but frames return as taubah, not failure.	Istiqamah — 'The most beloved deeds to Allah are those done consistently, even if small.' (Bukhari)	Fully private. Only user sees their streak. No leaderboard.
Khatam Together	Community Quran reading challenge. Each member marks daily pages. Collective progress bar visible to community. Monthly cycle. Community celebrates each khatam.	Jemaah Quran reading. Community connection through shared ibadah.	Individual pages private. Collective % visible to community.
Ilmu Quest	Daily 3-minute Islamic knowledge module: 1 ayat + tafsir, 1 hadith, 1 fiqh, 1 sejarah. 3-question quiz. Points for correct answers. Monthly Ilmu Badges.	Talab al-ilm — 'Seeking knowledge is obligatory upon every Muslim.' (Ibn Majah)	Points private. Ilmu Badges visible on profile (optional).
Amal Harian	Daily checklist: Subuh jemaah, 1 page Quran, 100 istighfar, helped someone, optional fasting. Glowing checkbox on completion. Week summary as private muhasabah.	Muhasabah — 'Account yourselves before you are accounted.' (Umar r.a.)	Fully private. Never shared publicly.
Jemaah Points	QR scan at mosque entrance for each solat/program. Points accumulated. Monthly recognition for most consistent attendance. Not ranked publicly — only shown as participation tier.	27x reward for jemaah prayer. Points remind of the reward without creating competition.	Points visible to user and AJK. No public ranking.

Feature	Mechanic	Islamic Parallel	Privacy
Kebun Amal	Anonymous community sadaqah garden. Each donation plants a virtual tree. Garden grows as community gives. No amounts or names shown — only the collective garden.	'Sadaqah does not decrease wealth.' (Muslim). Anonymity preserves ikhlas.	Fully anonymous. Amount and identity never shown.
Ramadan Mode	Special 30-day activation. Daily: iftar doa log, tarawih check-in, Quran log, sahur reminder. Community collective stats at Eid. Shareable community achievement card.	Ramadan — 'Whoever fasts Ramadan with faith and expecting reward, his previous sins are forgiven.' (Bukhari)	Individual logs private. Community totals public.

7. Use Cases

The system supports five primary actor types: AJK Admin (mosque committee administrator), Imam/Bilal (prayer leader/muezzin), Jemaah Member (mosque community member), Naqib/Naqibah (usrah group leader), and System Administrator. Each use case below occupies a dedicated page per the document standard.

UC-01: AJK Admin Manages Imam & Bilal Jadual

Use Case ID	UC-01
Use Case Name	AJK Admin Creates and Manages Imam & Bilal Prayer Schedule
Actor(s)	AJK Admin (primary). Imam / Bilal (notification recipient).
Preconditions	1. Mosque is subscribed to SAJDA (any tier). 2. AJK Admin is authenticated with Admin role. 3. At least one imam and one bilal are registered in the mosque's user directory.
Trigger	AJK Admin navigates to 'Jadual Imam & Bilal' from the mosque admin dashboard.
Main Flow	1. System displays the current month's jadual in a calendar grid view (5 prayers × 30 days). 2. Admin selects 'Buat Jadual Baru' or selects an existing week to edit. 3. System presents rotation template options: Manual Assignment, Weekly Rotation (auto-assign from registered imams in order), or Copy from Previous Week. 4. Admin selects Weekly Rotation. System auto-generates assignments from the registered imam list. 5. Admin reviews generated schedule. System highlights any conflicts (same person assigned twice in one day across different prayers). 6. Admin resolves conflicts by dragging assignments or selecting replacement from dropdown. 7. Admin clicks 'Simpan & Hantar Notifikasi'. 8. System saves jadual to database and sends push notifications to all assigned imams and bilals: 'Jadual solat bulan [Month] telah ditetapkan. Sila semak jadual anda.' 9. System sends individual daily reminder notifications to imam/bilal assigned for that day, 2 hours before solat Subuh.
Alternative Flow A — Imam Requests Leave	At any point: Imam submits leave request for specific prayer(s) via their app. System notifies AJK of the request with date/prayer and list of available replacements. AJK approves and selects replacement. System notifies original imam (leave approved) and replacement imam (new assignment).
Alternative Flow B — Last-Minute Change	AJK Admin edits an existing assignment within 3 hours of prayer time. System sends urgent push notification to affected imam and replacement. Change is logged with timestamp and editor identity for accountability records.
Post-conditions	Jadual saved and all assigned personnel notified. Monthly jadual visible to all mosque members on the prayer times screen ('Imam: [Name]' displayed with each prayer time).
Business Rules	BR-01: No imam may be assigned to more than 3 consecutive same-prayer slots without a rest day. BR-02: Friday Jumaat imam must be specifically designated — cannot be auto-assigned from general rotation. BR-03: All jadual changes must be logged with editor name and timestamp.
Non-Functional	Jadual page must load within 2 seconds. Notifications must be delivered within 60 seconds of scheduling.

UC-02: Jemaah Member Follows a Mosque

Use Case ID	UC-02
Use Case Name	Jemaah Member Registers and Follows a Subscribed Mosque
Actor(s)	Jemaah Member (unauthenticated initially)
Preconditions	Target mosque is subscribed to SAJDA. User has a smartphone with browser access.
Trigger	User hears about SAJDA from a mosque announcement, friend referral, or mosque poster with QR code.
Main Flow	1. User scans mosque QR code OR searches mosque name at sajda.my. 2. System displays mosque profile: name, address, prayer times, current programs, member count. 3. User taps 'Ikut Masjid Ini' (Follow This Mosque). 4. System prompts lightweight registration: phone number OR email, full name, preferred language (BM/EN). 5. System sends OTP to phone/email for verification. 6. User verifies OTP. Account created. Mosque following relationship established. 7. System presents onboarding: 'Pilih masjid utama anda' (if following multiple) and notification preferences. 8. User lands on home screen showing: today's prayer times with imam names, upcoming programs, community feed, and 'Mulakan Perjalanan' (Start Your Journey) gamification prompt. 9. System sends welcome notification: 'Selamat datang ke komuniti [Mosque Name]!'
Alternative Flow A — Multiple Mosque Following	User may follow additional mosques by searching at any time from the Discover tab. Each followed mosque appears in the Mosques tab. Primary mosque determines home screen prayer times. User receives program notifications from all followed mosques.
Alternative Flow B — Mosque Not Yet on SAJDA	User searches for a mosque not yet subscribed. System shows: 'Masjid ini belum dalam SAJDA. Mahu jemput masjid anda?' User can submit an invitation email to the mosque's AJK directly from the app. This creates a warm lead pipeline for SAJDA's mosque acquisition.
Post-conditions	User account created. Mosque following relationship stored. User receives prayer time notifications, program reminders, and community feed updates from followed mosque(s).
Business Rules	BR-04: Users may follow unlimited mosques but must designate exactly one as primary. BR-05: Mosque following is public at the aggregate level (member count shown) but individual members are not publicly listed without consent.
Non-Functional	Registration flow must complete in under 90 seconds on first use. OTP delivery must occur within 30 seconds.

UC-03: Jemaah Member Registers for a Program/Kelas

Use Case ID	UC-03
Use Case Name	Jemaah Member Discovers and Registers for a Mosque Program
Actor(s)	Jemaah Member (authenticated)
Preconditions	User is authenticated and following at least one mosque. The mosque has created at least one upcoming program.
Trigger	User sees a program notification or browses the Programs tab.
Main Flow	1. User opens SAJDA and navigates to 'Program' tab or taps a push notification about a program. 2. System displays program detail: title, date/time, venue, speaker/facilitator, description, capacity remaining, and registration status. 3. User taps 'Daftar Sekarang' (Register Now). 4. System confirms registration, deducts one from available capacity. 5. System sends confirmation notification: 'Pendaftaran berjaya! [Program Name] pada [Date/Time].' 6. System schedules reminder notifications: 24 hours before and 1 hour before the program. 7. On the day of the program, user receives check-in QR code in the app. 8. At the program, AJK scans user's QR or user self-checks-in to record attendance. 9. Post-program: system prompts brief feedback (1–5 stars + optional comment). Feedback visible only to AJK.
Alternative Flow A — Program Full	3a. Capacity is 0 (program full). 3b. System offers waitlist option. 3c. User joins waitlist. If a registered user cancels, the next person on the waitlist receives automatic notification and 2-hour window to confirm.
Alternative Flow B — Program Cancelled	AJK cancels a program after registrations have been collected. System notifies all registered users immediately. Cancellation reason (optional) displayed. AJK can offer alternative program suggestion.
Post-conditions	Registration stored. Attendance tracked post-program. Jemaah Points awarded for attended programs (if gamification enabled). AJK sees attendance statistics in dashboard.
Business Rules	BR-06: Registered users can cancel up to 2 hours before the program without penalty. BR-07: Programs with mandatory attendance (e.g., kelas Quran with progression) require AJK confirmation of each attendance. BR-08: Program feedback is anonymous to other members; visible to AJK only.
Non-Functional	Registration must complete within 3 taps and 5 seconds. Push notifications must deliver within 60 seconds of trigger.

UC-04: Jemaah Member Participates in Khatam Together Challenge

Use Case ID	UC-04
Use Case Name	Community Quran Khatam Together Challenge Participation
Actor(s)	Jemaah Member (authenticated). AJK Admin (challenge initiator).
Preconditions	AJK has activated the Khatam Together feature for the mosque (Kariah tier or above). User is a member of the mosque community.
Trigger	AJK starts a new Khatam cycle OR user joins an ongoing cycle.
Main Flow	1. AJK Admin navigates to Gamification > Khatam Together and selects 'Mulakan Khatam Baru'. 2. System creates a new khatam cycle: 30-juz divided across participating members, or open contribution model (each member contributes as much as they can). 3. All mosque members receive notification: 'Komuniti kita memulakan Khatam Quran bersama! Sertai sekarang.' 4. User joins the khatam cycle from the Quran tab. System shows: total juz remaining, community progress bar (% completed), days elapsed, estimated completion date. 5. User reads Quran (physical or digital) and logs daily contribution: tap the surah/ayat they reached today. 6. System updates community progress bar. If contribution is significant: 'JazakAllah! [Anonymous contribution] telah menambah X muka surat hari ini.' 7. When community reaches 100% (30 juz completed): system triggers khatam celebration notification to all members. 8. Community khatam count increments on mosque profile. Shareable celebration card generated. 9. AJK can initiate a new cycle immediately or schedule the next one.
Alternative Flow A — Ramadan Special Khatam	During Ramadan Mode, the Khatam challenge is enhanced: daily juz target displayed, tarawih readings integrated, completion by 27th Ramadan as community goal.
Gamification Notes	Individual daily contributions are private — other members see only that 'someone contributed X pages today' with no name attached. Collective milestone celebrations: 25%, 50%, 75%, 100% completion trigger community notifications. The Ikhlas Test is passed: no individual is ranked or celebrated; only the community achievement is celebrated.
Post-conditions	Community progress updated. Khatam count on mosque profile incremented on completion. Individual contribution logs stored privately for user's personal muhasabah only.
Business Rules	BR-09: Individual Quran contribution data is never shared with other users or shown on any leaderboard. BR-10: Khatam cycle minimum duration is 7 days. Maximum is 60 days before AJK must restart or close.

Non-Functional

Progress bar update must reflect within 5 minutes of user logging contribution.
Celebration notification must deliver to all members within 2 minutes of khatam completion.

UC-05: Jemaah Member Submits Keperluan Komuniti

Use Case ID	UC-05
Use Case Name	Community Member Submits a Need via Keperluan Komuniti Board
Actor(s)	Jemaah Member (authenticated — submitter). AJK Admin (moderator). Helper community member (responder).
Preconditions	Mosque has Kariah or Komuniti tier subscription. User is authenticated member of the mosque.
Trigger	A community member has a genuine need and submits it through the app.
Main Flow	1. User navigates to Keperluan Komuniti tab. 2. User taps 'Hantar Keperluan' and selects category: Bantuan Makanan, Bantuan Kewangan, Bantuan Perubatan, Keperluan Rumah, Kerja/Peluang, or Lain-lain. 3. User fills in: title (brief description), details (nature of need), urgency level (segera/tidak segera), and preferred contact method (through AJK only — their personal contact is never shown). 4. User submits. System notifies AJK: 'Keperluan baru untuk disemak.' 5. AJK reviews submission for authenticity and appropriateness. AJK can edit for privacy (remove identifying details) or approve as-is. 6. AJK approves. Need published to community board with category badge and anonymised description. 7. Community members who can help tap 'Saya Boleh Bantu'. System notifies AJK with helper's contact. 8. AJK facilitates connection between helper and person in need privately. 9. Once resolved, AJK marks need as 'Diselesaikan' (Resolved). Community sees '+1 keperluan diselesaikan oleh komuniti kita' without identifying parties.
Alternative Flow A — Urgent Medical Need	User selects urgency 'Segera' and category 'Bantuan Perubatan'. System immediately notifies AJK with urgent flag. AJK has option to broadcast to community members with medical volunteer registration. Response SLA displayed to submitter: 'AJK akan menghubungi dalam 2 jam.'
Privacy Architecture	The submitter's identity is never published on the board. Helpers contact AJK, not the person in need directly. AJK acts as privacy intermediary for all matching. Resolved needs show only the category and resolution — never the parties involved.
Post-conditions	Need published on board (if approved). Community members notified. Helper-need matching facilitated through AJK. Resolution tracked for mosque welfare statistics.
Business Rules	BR-11: AJK must review and approve/reject all submissions within 24 hours of submission. BR-12: Needs that involve financial amounts are never specified publicly — only 'bantuan kewangan diperlukan' without amount. BR-13: All resolved needs are archived (not deleted) for mosque welfare reporting to MAIS/JAKIM.

Non-Functional	Submission flow must complete in under 3 minutes. AJK notification must deliver immediately upon submission.
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UC-06: Jemaah Member Uses Ramadan Mode

Use Case ID	UC-06
Use Case Name	Community Participates in Ramadan Mode 30-Day Challenge
Actor(s)	All authenticated mosque members. AJK Admin (activator).
Preconditions	Mosque has Kariah or above tier. AJK has activated Ramadan Mode (auto-activates on 1 Ramadan if AJK has not manually disabled).
Trigger	First day of Ramadan. System auto-activates Ramadan Mode for all Kariah+ tier mosques.
Main Flow	- On 1 Ramadan, SAJDA home screen transforms: special Ramadan UI with moon/stars aesthetic, Ramadan countdown, and 'Ramadan Challenge' banner. - All mosque members receive notification: 'Ramadan Mubarak! Komuniti kita memulakan Ramadan Challenge bersama. Jom!' - User taps into Ramadan Challenge. Sees 30-day progress grid with today highlighted. - Each day contains 5 Ramadan activities to log (private): - Sahur reminder acknowledged (tap to confirm woke up for sahur) - Subuh jemaah check-in (QR or manual) - Quran reading (pages logged) - Tarawih check-in (QR at mosque) - Iftar doa read (tap to confirm) - Community screen shows collective stats (updated hourly, no individual data): - Total tarawih rakaat performed by community this Ramadan - Total Quran pages read by community - Number of community members active today - At Eid: system generates community achievement card with total stats. Shareable. Goes viral.
Viral Mechanic	Eid achievement card format: 'Komuniti Masjid [Name] pada Ramadan 1446H: ■ [X] rakaat tarawih bersama ■ [X] muka surat Quran dibaca ■ [X] ahli aktif sepanjang Ramadan Alhamdulillah — Ramadan kita penuh berkat.' This card is designed to be screenshot and shared on Instagram Stories, TikTok, and WhatsApp. It drives organic mosque-to-mosque spread every Ramadan — free marketing permanently.
Post-conditions	30-day Ramadan log stored privately per user for personal muhasabah. Community achievement stats published on mosque profile. Eid achievement card downloadable. Ramadan Mode deactivates automatically on 1 Syawal.
Business Rules	BR-14: All individual Ramadan activity logs are private and never shared without explicit user consent. BR-15: Community stats displayed are aggregate counts only — no individual names or amounts. BR-16: Ramadan Mode is the only feature that may be promoted externally with specific member counts — because it celebrates community, not individuals.

Non-Functional

Eid achievement card must generate within 5 seconds. Ramadan Mode activation must complete for all mosque members simultaneously within 1 minute of midnight on 1 Ramadan.

UC-07: Naqib/Naqibah Manages Usrah Group

Use Case ID	UC-07
Use Case Name	Usrah Group Leader Creates and Manages Their Tarbiyah Group
Actor(s)	Naqib/Naqibah (group leader). Usrah members. AJK Admin (usrah programme coordinator).
Preconditions	Mosque has Komuniti tier subscription. AJK has enabled the Usrah module and appointed Naqib/Naqibah.
Trigger	AJK Admin creates a usrah group and designates a Naqib/Naqibah, OR an existing Naqib manages their ongoing group.
Main Flow	1. AJK Admin creates usrah group: name, Naqib/Naqibah designation, target member count (5–12), meeting day/time, and tarbiyah syllabus reference. 2. Members are invited by Naqib via in-app invitation or direct AJK assignment. 3. Naqib logs each weekly meeting: date, location, attendance (tick/cross per member), topic covered, and brief notes. 4. System generates usrah attendance dashboard: individual consistency rates, group attendance trend, topics covered vs syllabus progress. 5. Naqib can assign weekly amal (good deeds) for members: 'Amal minggu ini: Solat Subuh berjemaah setiap hari.' 6. Members receive notification of weekly amal assignment. 7. Members self-report amal completion (private — Naqib sees completion rate, not which specific member completed). 8. AJK Admin sees aggregate usrah health dashboard: number of active usrah, overall attendance rates, members without usrah (potential for new group formation).
Alternative Flow A — Member Leaves Usrah	Member requests to leave or Naqib removes inactive member. System notifies AJK of change. Member is returned to 'Unassigned' pool for AJK to assign to another group.
Post-conditions	Weekly meeting logged. Attendance recorded. Amal assigned and tracked. AJK has full oversight of usrah programme health.
Business Rules	BR-17: Individual member usrah attendance records are visible only to that member, their Naqib, and AJK — never to other members. BR-18: Usrah groups are mosque-specific — members from different mosques cannot be in the same usrah group on SAJDA.
Non-Functional	Usrah meeting log must complete in under 5 minutes for a Naqib with 10 members. Amal notifications must deliver within 30 minutes of Naqib sending.

UC-08: AJK Admin Manages Digital Sadaqah Wallet

Use Case ID	UC-08
Use Case Name	AJK Admin Manages Mosque Digital Sadaqah Collection and Reporting
Actor(s)	AJK Admin (wallet manager). Jemaah members (donors). SAJDA Finance (transaction processing via Billplz).
Preconditions	Mosque has Kariah or Komuniti tier. AJK has completed mosque bank account verification with Billplz.
Trigger	AJK Admin activates sadaqah wallet OR a jemaah member initiates a donation.
Main Flow	1. AJK Admin sets up wallet: creates allocation categories, sets monthly collection target (optional), and publishes wallet to mosque community. 2. Community members see 'Sedekah Digital' section on their mosque home screen. 3. Member taps 'Beri Sedekah', selects allocation category, enters amount (minimum RM1), selects FPX or card. 4. Billplz processes payment. On success: SAJDA records transaction, deducts 1.5% fee (capped RM15), and credits mosque's Billplz collection account. 5. Donor receives confirmation: 'JazakAllah! Sedekah RM[X] berjaya. Semoga Allah melipatgandakan ganjaran kamu.' — no receipt needed; amount is their personal record. 6. Kebun Amal animation: a new virtual tree is planted in the anonymous community garden (no amount or name shown). 7. AJK Admin views real-time dashboard: total collected this month by category, total disbursed, running balance. 8. At month end, system auto-generates transparency report: total collected, total disbursed by category with purpose notes, balance. Report published to all mosque members.
Disbursement Flow	AJK Admin logs a disbursement: amount, category, recipient (anonymised as 'Keluarga 1', 'Pelajar 2'), purpose. Disbursement recorded in audit log. Monthly report includes all disbursements. Funds transferred to mosque bank account monthly by Billplz, minus SAJDA fee.
Post-conditions	Transaction recorded. Fee deducted. Mosque collection balance updated. Kebun Amal garden grows. Monthly report auto-generated. Full audit trail maintained.
Business Rules	BR-19: SAJDA charges 1.5% fee on all sadaqah transactions, minimum RM0.30, capped at RM15 per transaction. BR-20: All transactions are final — no refunds on sadaqah once confirmed (Islamic principle). BR-21: Mosque bank account must be verified (SSM-registered institution) before wallet activation. BR-22: Monthly transparency reports are mandatory — AJK cannot disable them.
Non-Functional	Payment processing must complete within 10 seconds. Transparency report must auto-generate on the 1st of each month by 8:00 AM. Dashboard must show real-time balance updated within 5 minutes of each transaction.

8. Functional Requirements

8.1 Mosque Administration Requirements

ID	Category	Requirement	Priority
FR-A01	Prayer Times	System shall display accurate daily prayer times for the subscribed mosque's location, calculated using Malaysia JAKIM parameters.	Must Have
FR-A02	Prayer Times	System shall display the assigned imam and bilal name alongside each prayer time.	Must Have
FR-A03	Imam Jadual	System shall support creating, editing, and publishing weekly imam/bilal rotation schedules.	Must Have
FR-A04	Imam Jadual	System shall detect and highlight assignment conflicts (same person, overlapping time).	Must Have
FR-A05	Imam Jadual	System shall send push notifications to assigned imams/bilals upon schedule publication and 2 hours before each assigned prayer.	Must Have
FR-A06	Imam Jadual	System shall support imam leave requests and replacement workflow with AJK approval.	Must Have
FR-A07	Programs	System shall support creating programs with title, description, date/time, venue, capacity, speaker, and registration requirement.	Must Have
FR-A08	Programs	System shall manage program capacity and automatically activate waitlist when full.	Must Have
FR-A09	Programs	System shall send reminder notifications to registered attendees 24 hours and 1 hour before program.	Must Have
FR-A10	Programs	System shall provide QR-based attendance check-in for registered programs.	Must Have
FR-A11	Announcements	System shall allow AJK to publish text and image announcements to all mosque followers.	Must Have
FR-A12	Keperluan	System shall provide a moderated community needs board with category filtering and AJK approval workflow.	Must Have
FR-A13	Volunteer	System shall support volunteer slot creation with skills/availability tagging and member signup.	Should Have
FR-A14	Dashboard	System shall provide AJK with a real-time dashboard: member count, program attendance trends, sadaqah collections, usrah health.	Must Have

8.2 Community Member Requirements

ID	Category	Requirement	Priority
FR-C01	Following	System shall allow authenticated users to follow one or more subscribed mosques.	Must Have
FR-C02	Following	System shall allow users to designate a primary mosque for home screen prayer times.	Must Have
FR-C03	Discovery	System shall provide mosque search by name, state, and location (geolocation-based).	Must Have
FR-C04	Programs	System shall allow jemaah to browse, register, and cancel program registrations.	Must Have
FR-C05	Community Feed	System shall display a structured community feed with categories: Pengumuman, Ilmu, Keperluan, Doa, Peristiwa.	Must Have
FR-C06	Doa Network	System shall support anonymous doa request submission and community 'Aamiin' responses.	Must Have
FR-C07	Sadaqah	System shall allow authenticated users to donate to their followed mosque's sadaqah wallet via FPX or card.	Must Have
FR-C08	Qiblat	System shall provide compass-based Qibla direction using device sensors.	Must Have
FR-C09	Hadis	System shall display a daily hadith from 40 Hadith Nawawi with AI-powered contextual explanation (Claude API).	Must Have
FR-C10	Notifications	System shall support configurable push notifications for prayer times, programs, community posts, and gamification milestones.	Must Have

8.3 Gamification Requirements

ID	Category	Requirement	Priority
FR-G01	Istiqamah	System shall track daily solat completion privately with visual streak indicator.	Must Have
FR-G02	Istiqamah	System shall frame missed streak as opportunity for taubah — no punishment language.	Must Have
FR-G03	Khatam	System shall support community Quran khatam cycles with collective progress bar.	Must Have
FR-G04	Khatam	System shall keep individual Quran contribution private; show only collective progress.	Must Have
FR-G05	Ilmu Quest	System shall deliver one daily Islamic knowledge module (ayat + hadith + fiqh + sejarah) with 3-question quiz.	Must Have
FR-G06	Ilmu Quest	System shall use Claude API to generate contextual explanations for daily hadith.	Must Have
FR-G07	Amal Harian	System shall provide a private daily good deeds checklist configurable by the user.	Must Have
FR-G08	Jemaah Points	System shall award points for mosque attendance via QR check-in, never displayed on public ranking.	Must Have
FR-G09	Kebun Amal	System shall display an anonymous community sadaqah garden — no amounts or names visible.	Must Have
FR-G10	Ramadan Mode	System shall auto-activate Ramadan Mode on 1 Ramadan for Kariah+ tier mosques.	Must Have
FR-G11	Ramadan Mode	System shall generate shareable Eid community achievement card with aggregate stats only.	Must Have
FR-G12	Privacy	System shall never display individual ibadah data publicly without explicit user opt-in.	Must Have

9. Non-Functional Requirements

ID	Category	Requirement	Target
NFR-01	Performance	Prayer time display page load on 4G mobile	< 2 seconds
NFR-02	Performance	Community feed initial load	< 2.5 seconds
NFR-03	Performance	QR check-in scan to confirmation	< 1 second
NFR-04	Availability	Prayer times feature uptime (safety-critical)	>= 99.9%
NFR-05	Availability	Overall platform uptime	>= 99.5%
NFR-06	Scalability	Concurrent mosque check-ins (Jumaat surge — 10,000 simultaneous)	Handled without degradation
NFR-07	Security	All data in transit	TLS 1.3
NFR-08	Security	B2B mosque data isolation	Supabase RLS on every table
NFR-09	Privacy	Individual ibadah data access	User-only by default; no shared access
NFR-10	Privacy	PDPA 2010 compliance for Malaysian users	Legal review confirmed
NFR-11	Islamic Compliance	Gamification features reviewed by Islamic advisor	Fatwa consultation documented
NFR-12	Cost	Claude API monthly spend cap	RM200/month (configurable)
NFR-13	Accessibility	Support for Malaysian Malay (BM) and English	Both languages on all screens
NFR-14	Mobile	PWA installable on Android 8+ and iOS 13+	No native app install required
NFR-15	Offline	Prayer times available offline (cached)	Yes — 24-hour cache

10. System Architecture

10.1 Architecture Overview

SAJDA is built on a modern serverless architecture optimised for a solo founder: zero infrastructure management overhead, automatic scaling, and predictable costs. The architecture separates the Mosque Admin SPA (desktop-optimised, complex admin workflows), the Community Member PWA (mobile-first, lightweight, offline-capable), and the REST API layer (shared between both clients and future third-party integrations).

10.2 Technology Stack

Layer	Technology	Justification
Frontend	Next.js 14 (App Router)	Server components for SEO on mosque discovery pages; client components for real-time community feed and gamification; single codebase for admin and member views.
Design System	Tailwind CSS + Custom Sacred Futurism tokens	Sacred green (#3DFF8F) accent on near-black (#0A1628) background; Syne (display) + DM Sans (body) + JetBrains Mono (data); BracketCard components; IslamicMesh SVG background pattern.
Database	Supabase (PostgreSQL 15)	Multi-tenant data isolation via RLS; Supabase Auth for all user types; Realtime for live community feed updates; Storage for program images and announcement media.
AI Layer	Claude API (claude-sonnet-4-6)	Daily hadith contextual explanation; Ilmu Quest question generation; Program description drafting for AJK; content moderation assistance.
Prayer Times	Aladhan.com API + JAKIM parameters	Malaysian calculation method (JAKIM); all 14 states supported; offline cache via Supabase Edge Functions.
Hadith API	fawazahmed0/hadith-api (GitHub)	Free, complete, multilingual hadith database; 40 Hadith Nawawi collection; BM translations available.
Billing	Billplz	Malaysian-native; FPX + card; recurring subscription API; webhook support for subscription lifecycle; sadaqah collection with disbursement to mosque bank accounts.
Push Notifications	Supabase + Web Push API	Browser push notifications via Service Worker; no third-party notification service needed; works on Android and iOS PWA.
Qibla Calculation	Device Compass API + Haversine formula	Client-side calculation using device magnetometer; no external API needed; works offline.
Hosting	Vercel Pro	Zero-config deployment; edge CDN for fast global delivery; preview deployments for QA.
Monitoring	Sentry + Vercel Analytics	Error tracking; Core Web Vitals monitoring; prayer time reliability alerts.

10.3 Multi-Tenant Architecture

SAJDA supports unlimited mosque tenants on a single Supabase project and Next.js application. Each mosque is a tenant identified by a `mosque_id` UUID. Every table in the mosque-scoped domain includes a `mosque_id` column. Row Level Security policies enforce that all queries return only rows matching the authenticated user's `mosque_id`. This means even a compromised API route cannot return data from a different mosque — the database enforces isolation at the row level.

The routing architecture uses middleware to route `sajda.my/[mosque-slug]` to the public mosque profile and `app.sajda.my/[mosque-slug]/dashboard` to the authenticated admin interface. Community member profiles are user-scoped (not mosque-scoped) — a single user account can follow multiple mosques and access each mosque's community features from their one account.

10.4 Offline Strategy

Prayer times are safety-critical — a user must be able to see prayer times even without internet. SAJDA implements a Progressive Web App Service Worker that caches prayer times for the current and next 7 days. The community feed, programs, and gamification features require connectivity and display graceful offline states with clear messaging. Qibla Finder works fully offline using the device compass and pre-cached mosque coordinates.

11. Database Schema

11.1 Core Tables

Table	Domain	Description & Key Columns
mosques	Mosque	Mosque tenant profiles. id, name, slug, address, state, coordinates, registration_number, subscription_plan, billing_status, created_at. RLS: public read for discovery; write restricted to AJK.
mosque_members	Mosque	AJK and assigned roles. id, mosque_id (FK), user_id (FK), role (ajk_admin imam bilal naqib naqibah), is_active, joined_at.
prayer_times	Mosque	Daily prayer times per mosque. id, mosque_id (FK), date, fajr, syuruk, zohor, asar, maghrib, isyak, created_at.
imam_jadual	Mosque	Prayer leader schedule. id, mosque_id (FK), imam_id (FK), prayer (fajr zohor asar maghrib isyak jumaat), scheduled_date, is_replacement, replacement_for (FK nullable), status (scheduled completed absent).
programs	Mosque	Events and classes. id, mosque_id (FK), title, description, program_type, start_datetime, end_datetime, venue, capacity, registration_required, speaker_name, status (upcoming ongoing completed cancelled), created_by.
program_registrations	Mosque	Member program signups. id, program_id (FK), user_id (FK), registered_at, attendance_status (registered attended absent waitlisted), feedback_rating (1–5).
announcements	Mosque	Mosque announcements. id, mosque_id (FK), title, body, image_url, published_at, expires_at, created_by.
community_needs	Mosque	Keperluan Komuniti. id, mosque_id (FK), category, title, description, urgency, status (pending approved in_progress resolved), submitted_by_hash, approved_by, created_at.
volunteer_slots	Mosque	Volunteer opportunities. id, mosque_id (FK), title, description, slots_available, slots_filled, date_time, required_skills.
usrah_groups	Community	Study groups. id, mosque_id (FK), name, naqib_id (FK), max_members, syllabus_ref, meeting_day, meeting_time, is_active.
usrah_members	Community	Usrah membership. id, usrah_id (FK), user_id (FK), joined_at, status (active inactive).
usrah_meetings	Community	Weekly meeting logs. id, usrah_id (FK), meeting_date, topic_covered, notes, logged_by. attendance tracked in usrah_attendance table.
user_following	User	Mosque following relationships. id, user_id (FK), mosque_id (FK), is_primary, followed_at.
user_profiles	User	Extended user profiles. id (FK to auth.users), full_name, phone_hash, preferred_language, notification_prefs (jsonb), onboarded_at.

Table	Domain	Description & Key Columns
ibadah_logs	Gamification	Private ibadah tracking (user-only RLS). id, user_id (FK), log_date, solat_fajr..solat_isyak (boolean), quran_pages (int), amal_completed (jsonb). NEVER accessible by mosque or other users.
community_khatam	Gamification	Community Quran progress. id, mosque_id (FK), cycle_name, start_date, end_date, total_pages_target, pages_completed, participant_count, status.
ilmu_modules	Gamification	Daily knowledge modules. id, date, ayat, tafsir_brief, hadith_text, hadith_source, fiqh_point, sejarah_snippet, quiz_questions (jsonb), created_by (system admin).
sadaqah_transactions	Sadaqah	Donation records. id, mosque_id (FK), donor_hash (one-way), amount, fee_deducted, net_amount, category, payment_ref (Billplz), status, created_at. RLS: AJK read, user read own.
sadaqah_disbursements	Sadaqah	Disbursement records. id, mosque_id (FK), amount, category, purpose, recipient_code (anonymised), disbursed_by, disbursed_at.
doa_requests	Community	Anonymous prayer requests. id, mosque_id (FK), category (illness career family exam general), aamiin_count (int), status (active resolved), created_at. No user linkage stored.
subscriptions	Billing	Mosque subscription records. id, mosque_id (FK), plan (surau kariah komuniti), billplz_subscription_id, status, current_period_start, current_period_end, trial_ends_at.

12. Security and Privacy Design

12.1 Islamic Privacy Principles

SAJDA's privacy architecture is informed by Islamic principles of privacy (hifz al-'ird — protecting honour) and sincerity (ikhlas). The most critical Islamic privacy principle in this platform is that individual ibadah must remain between the worshipper and Allah. No ibadah tracking data — solat logs, Quran pages, amal completion — is ever accessible to any other user, mosque staff, or administrator. This is enforced at the database level with strict RLS policies, not just application-level access controls.

12.2 Row Level Security — Critical Policies

Table	Read Policy	Write Policy
ibadah_logs	user_id = auth.uid() ONLY. No exceptions. Not accessible to AJK, imam, or admin.	user_id = auth.uid() ONLY.
doa_requests	mosque_id in user's followed mosques. No user identity stored.	Any authenticated follower of the mosque.
sadaqah_transactions	donor sees own transactions. AJK sees all mosque transactions. Other users: none.	Authenticated users only. System writes on payment confirmation.
community_needs	Approved needs visible to mosque followers. Pending: AJK only.	Authenticated mosque members. AJK for approval.
usrah_meetings	Naqib sees own usrah. AJK sees all. Members: none.	Naqib only.
imam_jadual	All mosque followers (read published schedule). AJK (full edit).	AJK Admin role only.
programs	All mosque followers (read). AJK (create/edit).	AJK Admin role only.

12.3 PDPA Compliance

- Personal data collected: name, phone number (hashed for ibadah logs), email. Minimal data principle applied.
- Donor identity: stored as one-way SHA-256 hash. The actual donor identity cannot be retrieved from stored data.
- Doa request submitters: zero identity data stored. Completely anonymous submissions.
- Community needs submitters: identity known to AJK only; never stored in the needs record itself.
- Data retention: active accounts retained for 3 years after last activity. Deletion request processed within 30 days per PDPA Section 30.
- Third-party processors: Billplz (payment), Vercel (hosting), Supabase (database), Anthropic (AI). All have appropriate data processing agreements.

13. Commercial Model

13.1 Subscription Tiers

Tier	Price	Target	Features
Surau	RM49/mo	Small surau, 50–200 jemaah	Prayer times, basic announcements, imam jadual, Hadis Harian, Qiblat Finder. Core functionality only.
Kariah	RM149/mo	Standard masjid kariah, 200–1,000 jemaah	All Surau features + Program management, Keperluan Komuniti, Community feed, Digital sadaqah wallet, All gamification features (Istiqamah, Khatam, Ilmu Quest, Ramadan Mode), Volunteer slots.
Komuniti	RM349/mo	Large masjid, musollas network, 1,000+ jemaah	All Kariah features + Usrah management, Islamic mentor matching, Cross-mosque challenges, Priority support, API access, Custom branding, Multi-branch support.
Enterprise	Custom	State religious departments, JAKIM, MAIS	Everything in Komuniti × multiple mosques. White-label option. Dedicated account manager. Custom integrations. Annual contract.

13.2 Revenue Streams

Stream	Model	Projection (Month 12)
Mosque subscriptions	Monthly recurring: RM49–349/mo per mosque	50 Surau + 30 Kariah + 10 Komuniti = RM2,450 + RM4,470 + RM3,490 = RM10,410 MRR
Sadaqah transaction fee	1.5% of all digital sadaqah collected (capped RM15/transaction)	90 mosques × RM10,000 avg collection/mo × 1.5% = RM13,500 MRR
Premium individual membership	RM9.90/mo for jemaah without a subscribed mosque (direct-to-consumer)	5,000 individual users × RM9.90 = RM49,500 MRR (Year 2 target)
Enterprise contracts	Custom annual contracts for MAIS, JAKIM, state religious departments	1 MAIS contract = RM2,000–5,000/mo (Year 2)

Month 12 MRR target: RM10,410 (subscription) + RM13,500 (sadaqah fee) = RM23,910/mo

14. Deployment Architecture

14.1 Environments

Environment	Platform	Purpose
Local Development	Next.js dev + Supabase local	Feature development; mock prayer times; no real payments
Preview	Vercel Preview per PR	Feature review; real Supabase staging; sandbox Billplz
Staging	Vercel + Supabase staging project	Pre-production testing; real Claude API (reduced limits); Billplz sandbox
Production	Vercel Pro + Supabase Pro	Live system; monitoring active; full RLS enforced; real payments

14.2 PWA Android Deployment

SAJDA targets Android via PWABuilder (Microsoft's free PWA-to-APK tool). The compiled APK is published on Google Play Store. This avoids the RM100+/year Apple Developer fee for the MVP phase while covering the majority of Malaysian smartphone users (Android market share >75% in Malaysia). iOS users access SAJDA via Safari PWA with 'Add to Home Screen' prompt. Native iOS app is a Phase 2 priority once revenue supports the Apple Developer account.

14.3 Cost Structure

Service	Usage	Monthly Cost
Vercel Pro	Next.js hosting + edge functions	RM94/mo
Supabase Pro	PostgreSQL + Auth + Storage + Realtime	RM47/mo
Claude API	Hadis context + Ilmu Quest + AJK content assist	RM50–200/mo (usage-based)
Aladhan API	Prayer times (Malaysian parameters)	RM0 (free tier)
fawazahmed0 Hadith API	Daily hadith data	RM0 (free, GitHub-hosted)
Billplz	Subscription billing + sadaqah processing	1.5% per sadaqah tx; RM0.30 per subscription billing
Resend	Transactional email	RM0 (free tier to 3,000/mo)
Domain	sajda.my + app.sajda.my	RM5/mo
Google Play Store	Android PWA distribution	RM100 one-time

15. Risk Register

ID	Risk	Like.	Impact	Mitigation
R-01	Islamic scholar objection to gamification features	Medium	High	Engage JAKIM/MAIS advisor before launch. Document ikhlas-first design rationale. Seek fatwa consultation.
R-02	Low mosque adoption — AJK resistance to new technology	High	High	Free 3-month trial. White-glove onboarding. Start with Masjid Saujana Utama as flagship reference. Testimonials drive trust.
R-03	Youth using app for points without genuine ibadah intention	Medium	Medium	Private-only ibadah tracking. No public individual rankings. Regular design audits against ikhlas principle.
R-04	Sadaqah misuse by unverified mosque AJK	Low	High	SSM verification required for wallet activation. Monthly transparency reports mandatory. Full audit trail maintained.
R-05	Prayer time inaccuracy causing community distrust	Low	Critical	Use JAKIM-certified calculation parameters. Fallback to direct JAKIM feed if Aladhan API unavailable. 99.9% uptime target.
R-06	Competition from Muslim Pro adding community features	Medium	Medium	SAJDA's moat is physical mosque integration — Muslim Pro has no mosque relationships. Data moat deepens with time.
R-07	WhatsApp group resistance — AJK prefer familiar tools	High	Medium	Position SAJDA as complement to WhatsApp, not replacement. SAJDA pushes notifications to WhatsApp via webhook in Phase 2.
R-08	PDPA compliance failure — ibadah data exposure	Low	Critical	Strictest RLS policies. Regular security audit. ibadah_logs accessible only to the owning user — no exceptions in code.
R-09	Solo founder bandwidth — too many features, nothing shipped	High	High	Strict MVP discipline. Phase 1 = core 8 features only. No gamification until Phase 2. Revenue before features.
R-10	Billplz regulatory requirement for sadaqah collection	Medium	High	Consult Billplz and BNM on mosque donation collection compliance. Engage Islamic finance lawyer before wallet launch.

16. Product Roadmap

Phase 1 — Core MVP (Weeks 1–8)

Goal: Get Masjid Saujana Utama paying. Prove the core value proposition.

1. Supabase multi-tenant schema migration. Next.js 14 project scaffold with Sacred Futurism design system.
2. Prayer times display with JAKIM parameters. Imam/bilal jadual creation and notification.
3. Program management: create, publish, register, attendance QR check-in.
4. Announcement publishing. Keperluan Komuniti board with AJK moderation workflow.
5. Mosque following and jemaah member PWA. Hadis Harian with Claude API context.
6. Qiblat Finder (compass-based). Push notifications via Web Push API.
7. Billplz subscription billing (Surau + Kariah tiers). AJK admin dashboard.
8. Pre-launch QA + PDPA review. Onboarding for Masjid Saujana Utama. First invoice.

Phase 2 — Community & Gamification (Month 3–6)

Goal: 20 paying mosques. 5,000 active jemaah members. Validate community features.

- Papan Komuniti community feed with categories and Islamic reactions (MashaAllah, BarakAllah, Aamiin).
- Doa request network — anonymous submission, community Aamiin response.
- Istiqamah Meter (private solat streak). Khatam Together community Quran challenge.
- Ilmu Quest — daily Islamic knowledge module with Claude API question generation.
- Amal Harian private daily checklist. Kebun Amal anonymous sadaqah garden.
- Digital sadaqah wallet with Billplz FPX/card integration. Monthly transparency reports.
- Ramadan Mode — activate 2 months before Ramadan for first Ramadan launch.
- Android APK via PWABuilder. Google Play Store publication.

Phase 3 — Depth & Scale (Month 6–12)

Goal: 100 paying mosques. RM23K MRR. Malaysia market dominance.

- Usrah group management (Komuniti tier). Naqib dashboard and weekly meeting logging.
- Islamic mentor matching — Cari Ilmu feature. Skills registry within mosque community.
- Cross-mosque challenge network. Inter-mosque Ramadan Khatam competition.
- Volunteer slot management with skills matching. Community welfare reporting.
- Komuniti tier launch. Enterprise tier proposal to MAIS Selangor.
- Indonesia market entry — Bahasa Indonesia interface, local payment gateway.
- JAKIM/MAIS partnership exploration for official mosque digital platform endorsement.

Phase 4 — Islamic OS Vision (Year 2–5)

The long-term vision: SAJDA as the complete digital operating system for the global Muslim community — from daily solat to financial inclusion to religious education to community welfare. Each product addition is both commercially valuable and spiritually purposeful. This is how SAJDA becomes the most important Islamic technology company to emerge from Malaysia.

- Islamic microfinance — qardhul hasan (interest-free community loans through mosque network).
- Waqf management — digital waqf registration, tracking, and beneficiary management.
- Zakat collection and distribution — official MAIS/LHDN integration.

- Islamic education platform — structured curriculum, certified online kelas, ustaz marketplace.
- Halal compliance tool — ingredient traceability for small food businesses in the mosque community.
- AmanahDoc integration — Wasiat and Faraid planner accessible through SAJDA family profile.
- Global expansion: Indonesia, Saudi Arabia, UAE, Egypt, United Kingdom (Muslim diaspora).

Appendix A: API Specification

A.1 Public Mosque Discovery API

Unauthenticated endpoints for mosque discovery (used by member PWA before login):

```
GET /api/v1/mosques/search?q={name}&state;={state}■={lat}&lmg;={lng}
```

Returns: mosque_id, name, slug, state, subscription_tier, jemaah_count, next_prayer_time.

```
GET /api/v1/mosques/{slug}/public
```

Returns: mosque profile, prayer times (today + 7 days), upcoming programs (public), announcement count.

A.2 Authenticated Community API

All endpoints require Bearer token (Supabase JWT).

```
GET /api/v1/user/feed – Community feed from all followed mosques
```

```
POST /api/v1/programs/{id}/register – Register for a program
```

```
GET /api/v1/gamification/today – Today's Ilmu Quest + Amal Harian
```

```
POST /api/v1/gamification/log – Log ibadah (private, user-only RLS)
```

```
POST /api/v1/sadaqah/donate – Initiate sadaqah payment via Billplz
```

```
GET /api/v1/khatam/{mosque_id} – Community khatam progress
```

A.3 AJK Admin API

Requires AJK Admin role JWT claim.

```
POST /api/v1/admin/jadual – Create/update imam jadual
```

```
POST /api/v1/admin/programs – Create program
```

```
GET /api/v1/admin/dashboard – AJK metrics dashboard data
```

```
GET /api/v1/admin/sadaqah/report – Monthly sadaqah transparency report
```

```
POST /api/v1/admin/needs/{id}/approve – Approve keperluan komuniti submission
```


Appendix B: Islamic & Technical Glossary

Term	Definition
Amal Jariah	Continuous charity — good deeds whose reward continues after death, often used to describe the lasting impact of building mosques, sharing knowledge, or raising righteous children.
Berlumba-lumba dalam Kebaikan	Competing in goodness — Quranic principle (2:148) that Muslims should compete with each other in performing good deeds, not in worldly status.
Fardu Ain	Individual obligation — Islamic duties that every adult Muslim must personally perform, such as the five daily prayers.
Fardu Kifayah	Collective obligation — Islamic duties that must be performed by the community but not necessarily every individual, such as funeral prayers.
Gamification	The application of game-design elements (points, streaks, challenges, rewards) to non-game contexts to increase engagement and motivation.
Hifz al-'Ird	Protection of honour — one of the five Islamic objectives (maqasid al-shariah) that includes protecting individual dignity and privacy.
Istiqamah	Steadfastness and consistency in righteous deeds. 'The most beloved deed to Allah is the most regular and constant, even if it were little.' (Bukhari 6465)
Ikhlas	Sincerity of intention — performing acts of worship solely for the sake of Allah, without desire for recognition or worldly reward.
Khatam	Completion — specifically the completion of reciting the entire Quran (30 juz) from Al-Fatiha to An-Nas.
Muhasabah	Self-reckoning and accountability — the Islamic practice of reflecting on one's deeds, intentions, and character for spiritual growth.
Naqib/Naqibah	Male/female leader of a tarbiyah (spiritual development) group, responsible for guiding members' religious growth and accountability.
Qardhul Hasan	Interest-free loan — an Islamic financial instrument where a lender provides funds without charging interest, expecting only the principal back.
Riak	Showing off — the sin of performing acts of worship or good deeds to be seen and praised by others rather than for Allah's pleasure.
Sacred Futurism	SAJDA's design aesthetic: the visual language of Islam's timeless principles expressed through modern, digital-native design. Sacred green (#3DFF8F) on near-black backgrounds, Islamic geometric patterns as digital motifs.
Tarbiyah	Islamic upbringing and character formation — the systematic process of nurturing Islamic values, knowledge, and practice in an individual.
Waqf	Islamic endowment — the donation of assets (property, cash, or shares) to a charitable cause in perpetuity, with ongoing rewards for the donor.

Term	Definition
Wird	A daily litany of Quranic verses, prayers, and remembrances that a Muslim maintains as a consistent spiritual practice.

Appendix C: References

Regulatory & Religious References

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Appendix D: Document Sign-Off

Document Revision History

Ver.	Date	Author	Description
1.0	April 2026	Muhamad Danish Farhan bin Zailan	Initial release — full SRS including enhanced gamification, community features, 8 detailed use cases, complete data model, commercial model, and 4-phase roadmap.

Approval Sign-Off

Role	Full Name	Signature	Date
Founder & Lead Engineer	Muhamad Danish Farhan bin Zailan		
Islamic Advisor / Consultant			
Technical Reviewer			
Legal Reviewer (PDPA)			
Business Stakeholder			

— End of Document —

SAJDA — Islamic Community Platform | SRS v1.0 | Muhamad Danish Farhan bin Zailan | April 2026

"And whoever does righteous deeds, whether male or female, while being a believer — those will enter Paradise and will not be wronged, [even as much as] the speck on a date seed." (An-Nisa 4:124)